

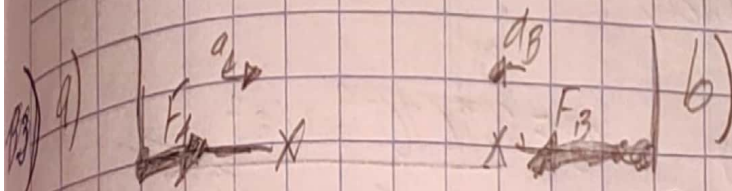
Gravitación

Ejercicio 13.7 Log

$$F = G \frac{m_1 m_2}{r^2} \quad S = \text{sol} \quad L = 1 \text{ um}$$

$$\frac{F_E}{F_S} = \frac{G \left(\frac{m_E m_L}{r_{EL}^2} \right)}{G \left(\frac{m_E m_S}{r_{ES}^2} \right)} = \frac{m_L}{m_S} \left(\frac{r_{ES}}{r_{EL}} \right)^2$$

$$= \frac{(7.99 \times 10^{30} \text{ Kg})}{(5.98 \times 10^{24} \text{ Kg})} \left(\frac{3.84 \times 10^8 \text{ m}}{1.5 \times 10^{11} \text{ m}} \right)^2 = 2.18$$



$$F_A = m_A a_A \quad a_A = \frac{F_A}{m_A}$$

$$F_A = F_B = G \frac{m_A m_B}{r^2} = (6.67 \times 10^{-11} \text{ N m}^2 / \text{Kg}^2) \frac{(15 \text{ Kg})(72 \text{ Kg})}{(120 \text{ m})^2}$$

$$= 7.807 \times 10^{-10} \text{ N}$$

$$a_A = \frac{7.807 \times 10^{-10}}{15 \text{ Kg}} = 5.2 \times 10^{-11} \text{ m/s}^2$$

$$a_B = \frac{7.807 \times 10^{-10}}{72 \text{ Kg}} = 1.1 \times 10^{-11} \text{ m/s}^2$$

$$x = x_0 + v_0 t + \frac{1}{2} a t^2$$

$$x_B = \frac{1}{2} a_B t^2 \quad x_A + x_B = 20 \text{ m} \quad 20 \text{ m} = \frac{1}{2} (a_A + a_B) t^2$$

$$t = \sqrt{\frac{2 \times 20}{5.2 \times 10^{-11} + 1.1 \times 10^{-11}}} = 7.32 \times 10^6 \text{ s} = 15 \text{ días}$$

$$13.5) F_A = G \frac{m_A m_B}{r^2} = G \frac{(0.26 \text{ kg})(0.01 \text{ kg})}{(0.100 \text{ m})^2} = 1.735 \times 10^{-11} \text{ N}$$

$$F_B = G \frac{m_B m}{r^2} = 1.735 \times 10^{-11} \text{ N}$$

$$F_{Ax} = -F_A \sin \theta = -(1.735 \times 10^{-11} \text{ N})(0.8) = -1.39 \times 10^{-11} \text{ N}$$

$$F_{Ay} = F_A \cos \theta = (1.735 \times 10^{-11})(0.6) = 1.04 \times 10^{-11} \text{ N}$$

$$F_{Bx} = F_B \sin \theta = 1.39 \times 10^{-11} \text{ N}$$

$$F_{By} = F_B \cos \theta = 1.04 \times 10^{-11}$$

$$\Sigma F_x = m a_x \rightarrow F_{Ax} + F_{Bx} = m a_x$$

$$2(1.04 \times 10^{-11}) = (0.01 \text{ kg}) a_x \quad a_x = 2.1 \times 10^{-9} \text{ m/s}^2$$

$$13.7) a) F_{\text{univ}} = F_g = (6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2) \frac{(1.35 \times 10^{22} \text{ kg})(70)}{(3.75 \times 10^8)^2} = 2.4 \times 10^{-9} \text{ N}$$

$$b) F_{\text{tierra}} = W = (70 \text{ kg})(9.8) = 690 \text{ N} \quad F_{\text{univ}}/F_{\text{tierra}} = 3.5 \times 10^{-6}$$

$$13.9) a) F_x = F_{1x} + F_{2x} = -G \frac{3 \text{ ml}}{x^2} + G \frac{\text{ml}}{(1 \text{ m} - x)^2} =$$

$$0.3(1 \text{ m} - x)^2 = x^2 \quad 1 \text{ m} - x = \frac{x}{\sqrt{3}} \quad x = \frac{1 \text{ m}}{1 + \sqrt{3}} = 0.634 \text{ m}$$

13.71) La distancia desde el centro de la tierra debe aumentarse en un factor de $\sqrt{10}$ entonces =

$$(\sqrt{10} - 1) R_{\text{tierra}} = 1.38 \times 10^4 \text{ m}$$

$$a) m_T = m_E / 1700 \quad R_T = \frac{R_E}{48}$$

$$= \frac{G \left(\frac{m_E}{1700} \right)}{\left(\frac{R_E}{48} \right)^2} = \left(\frac{609}{1700} \right) \frac{G m_E}{R_E^2} = 0.0377 g_E$$

$$a_T = (0.0377)(9.8) = \underline{0.37 \text{ m/s}^2}$$

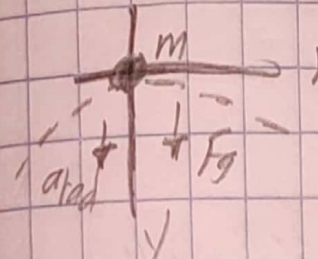
$$b) \rho_T = \frac{m_T}{\frac{4}{3}\pi R_T^3} = \frac{\frac{m_E}{1700}}{\frac{4}{3}\pi \left(\frac{R_E}{48} \right)^3} = \frac{5.72}{1700} \rho_E = \frac{5.72}{1700} (5500) = \underline{1700 \text{ kg/m}^3}$$

$$c) F_g = G \frac{m m_E}{r^2} \quad r = 600 \times 10^3 \text{ m} + R_E \quad F_g = 670 \text{ N}$$

$$W = mg = 735 \text{ N}$$

$$d) a) v = \sqrt{\frac{2(6.673 \times 10^{-11})(6.42 \times 10^{23})}{3.4 \times 10^{10}}} = \underline{5.02 \times 10^3 \text{ m/s}}$$

$$b) v = \sqrt{\frac{2(6.673 \times 10^{-11})(7.9 \times 10^{27})}{6.91 \times 10^7}} = \underline{6.06 \times 10^4 \text{ m/s}}$$

a) 

$$\sum F_g = m a_y \quad F_g = m a_{\text{rad}} \quad G \frac{m m_E}{r^2} = m \frac{v^2}{r}$$

$$v = \sqrt{\frac{G m_E}{r}} = \sqrt{\frac{(6.673 \times 10^{-11})(5.97 \times 10^{24})}{7.76 \times 10^6}}$$

$$v = \underline{7.46 \times 10^3 \text{ m/s}}$$

$$b) T = \frac{2\pi r}{v} = \frac{2\pi (7.76 \times 10^6)}{7.46 \times 10^3} = 6030 \text{ s} = \underline{7.68 \text{ hours}}$$

Ejercicio Sección 14.1 Descripción de la oscilación

14.1) a) $T = \frac{1}{f} = \frac{1}{400 \text{ Hz}} = 2.5 \times 10^{-3} \text{ s}$

$\omega = 2\pi f = 2\pi(400) = 2.51 \times 10^3 \text{ rad/s}$

b) $f = \frac{1}{T} = \frac{1}{50 \times 10^{-6} \text{ s}} = 2 \times 10^4 \text{ Hz}$

$\omega = 2\pi f = 2\pi(2 \times 10^4) = 1.26 \times 10^5 \text{ rad/s}$

c) $f = \frac{\omega}{2\pi} = \frac{2.7 \times 10^{15}}{2\pi} = 4.3 \times 10^{14} \text{ Hz}$

$\frac{4.3 \times 10^{15} \text{ rad/s}}{2\pi} = 7.5 \times 10^{14} \text{ Hz}$ $T = \frac{1}{f} = \frac{1}{7.5 \times 10^{14}} = 1.3 \times 10^{-15} \text{ s}$

$\frac{1}{4.3 \times 10^{14} \text{ Hz}} = 2.3 \times 10^{-15} \text{ s}$

d) $T = \frac{1}{f} = \frac{1}{5 \times 10^6 \text{ Hz}} = 2 \times 10^{-7} \text{ s}$ $\omega = 2\pi f = 2\pi(5 \times 10^6) = 3.14 \times 10^7 \text{ rad/s}$

14.3) $\text{Periodo} = \frac{0.5 \text{ s}}{1140} = 4.38 \times 10^{-4} \text{ s}$

Frecuencia angular $\omega = \frac{2\pi}{T} = 5.53 \times 10^3 \text{ rad/s}$

14.5) $T = \frac{1}{f} = 0.200 \text{ s}$ $t = 0.0500 \text{ s}$

14.7) a) $T = \frac{1}{f} = \frac{1}{6 \text{ Hz}} = 0.167 \text{ s}$

b) $\omega = 2\pi f = 2\pi(6) = 37.7 \text{ rad/s}$

c) $\omega = \sqrt{\frac{k}{m}} = m = \frac{k}{\omega^2} = \frac{120 \text{ N/m}}{(37.7 \text{ rad/s})^2} = 0.0844 \text{ kg}$

$$X = 0.320 \text{ m}$$

$$t_1 = 0$$

$$X = 0.160 \text{ m}$$

$$0.160 \text{ m} = (0.320 \text{ m}) \cos(\omega t_1)$$

$$\cos(\omega t_1) = 0.500$$

$$\omega t_1 = 1.047 \text{ rad}$$

$$t_1 = \frac{1.047}{6.981} = 0.150 \text{ s}$$

$$t_1 - t_2 = 0.150 \text{ s}$$

$$X = 0$$

$$t_2 = \frac{1.571}{6.981} = 0.225 \text{ s}$$

$$t_2 - t_1 = 0.225 - 0.150 = 0.0750 \text{ s}$$

$$A = \left| \frac{V_x}{\omega} \right| = \left| \frac{V_0}{\omega} \right| = 0.98 \text{ m}$$

$$X(0) = 0$$

$$\theta = \frac{\pi}{2} \quad V_{0x} < 0$$

$$X = (-0.98 \text{ m}) \sin((12.2) t)$$

$$a_c = -(2\pi(2.54 \text{ Hz}))^2 (1.7 \times 10^{-2} \text{ m}) = -2.71 \text{ m/s}^2$$

$$\omega = 15.7 \text{ rad/s} \quad X = (7.46 \text{ cm}) \cos((15.7 \text{ rad/s}) t + 0.715 \text{ rad})$$

$$V_x = (-22.9 \text{ cm/s}) \sin((15.7 \text{ rad/s}) t + 0.715)$$

$$a_x = (-359 \text{ cm/s}^2) \cos((15.7 \text{ rad/s}) t + 0.715 \text{ rad})$$

$$T = 2\pi \sqrt{\frac{m}{K}} \quad K = \frac{4\pi^2 m}{T^2} = \frac{4\pi^2 (42.5 \text{ kg})}{(7.35)^2} = 993 \text{ N/m}$$

$$T = 2\pi \sqrt{\frac{m}{K}} \quad m = \frac{T^2 K}{4\pi^2} = \frac{(7.35)^2 (993 \text{ N/m})}{4\pi^2} = 762 \text{ kg}$$

$$m_{\text{person}} = 762 \text{ kg} - 42.5 \text{ kg} = 720 \text{ kg}$$

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14.17) a) $T = 2\pi \sqrt{\frac{m}{K}}$ $m = K \left(\frac{T}{2\pi} \right)^2 = 250 \text{ N/m} \left(\frac{0.25}{2\pi} \right)^2 = 0.253 \text{ kg}$

b) $A = \frac{m a_{\text{max}}}{K} = \frac{(0.253 \text{ kg})(72 \text{ m/s}^2)}{250 \text{ N/m}} = 0.0727 \text{ m} = 7.27 \text{ cm}$

c) $F_{\text{max}} = KA = (250 \text{ N/m})(0.0727 \text{ m}) = 3.03 \text{ N}$

14.19) a) $T = \frac{2\pi}{\omega} = \frac{2\pi}{4.76 \text{ rad/s}} = 1.51 \text{ s}$

b) $\omega = \sqrt{\frac{K}{m}}$ $K = m\omega^2 = (7.5 \text{ kg})(4.76 \text{ rad/s})^2 = 26.1 \text{ N/m}$

c) $v_{\text{max}} = \omega A = (4.76 \text{ rad/s})(7.4 \text{ cm}) = 30.8 \text{ cm/s}$

d) $F_x = -Kx$ $F_{\text{max}} = KA = (26.1 \text{ N/m})(0.0740 \text{ m}) = 1.92 \text{ N}$

e) $x(t) = A \sin(\omega t + \phi)$ $t = 1 \text{ s}$ $x = -0.0125 \text{ m}$
 $v_x = -\omega A \cos(\omega t + \phi) = 30.4 \text{ cm/s}$

$a_x = -Kx/m = -\omega^2 x = 0.21 \text{ m/s}^2$

f) $F_x = -Kx = -(26.1 \text{ N/m})(-0.0125 \text{ m}) = 0.325 \text{ N}$